

RepurposeAlpha — Portfolio Analysis

Disease focus: Prader-Willi Syndrome

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Methodology

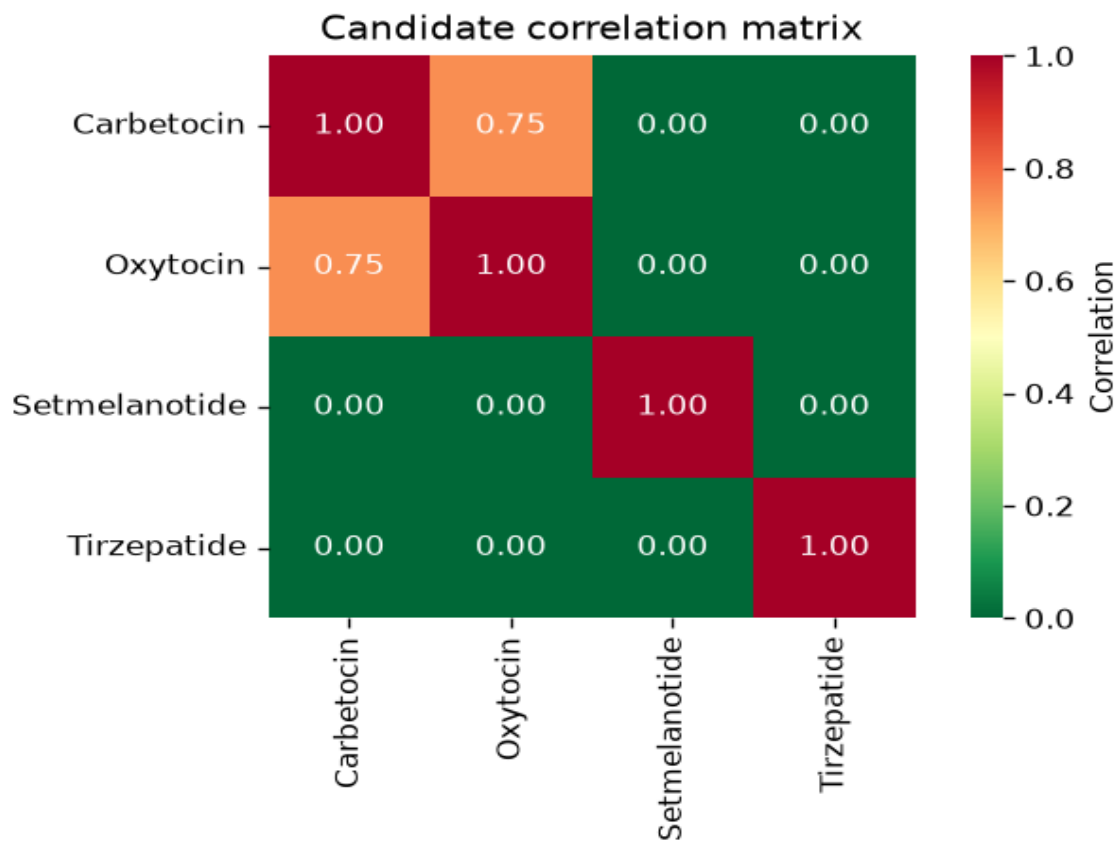
RepurposeAlpha applies Modern Portfolio Theory to drug repurposing. Candidate drugs are treated as correlated, risky assets. Correlations are computed from public bioactivity data (ChEMBL), chemical structure similarity, mechanism of action, and WHO ATC classification. Portfolio weights are selected by maximizing the Sharpe ratio subject to long-only constraints.

Key findings

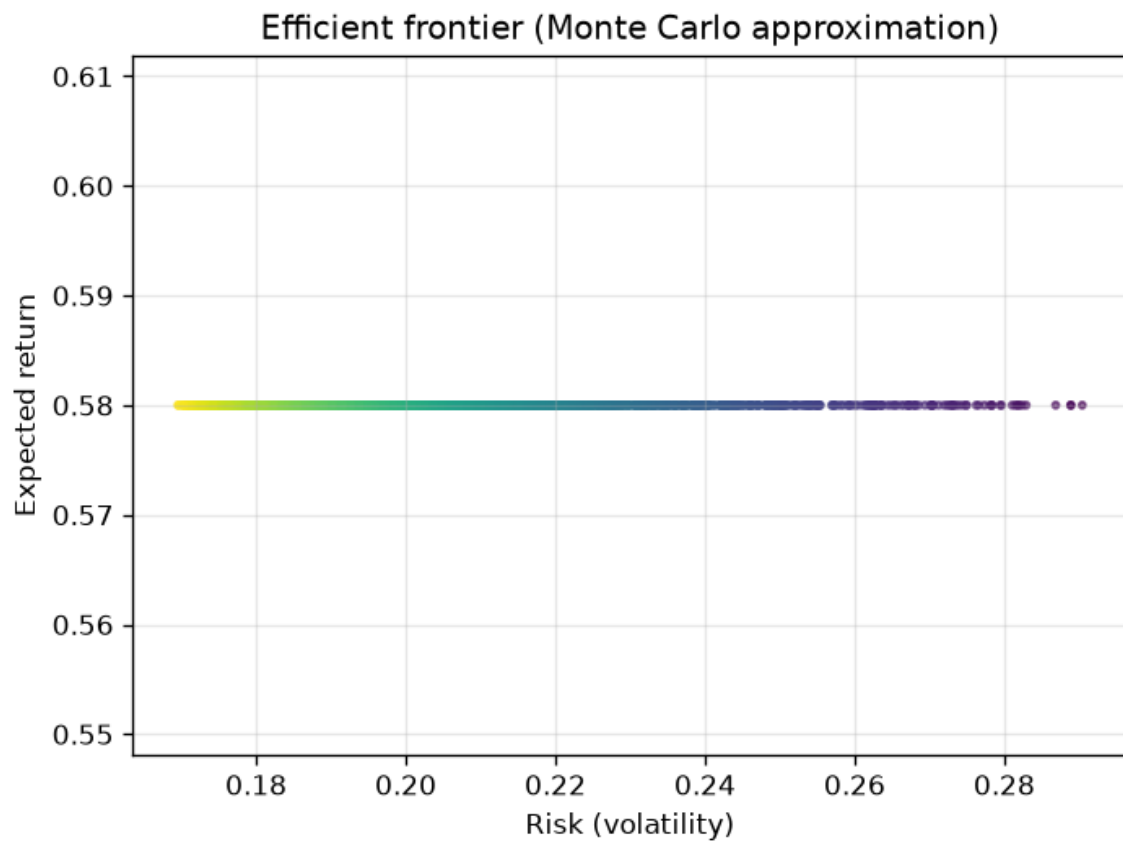
On the Prader-Willi Syndrome candidate set, the correlation engine identified a strong relationship between the two oxytocin-pathway drugs (Carbetocin and Oxytocin). The Markowitz optimizer recommends diversifying across the MC4R and GLP-1 pathways to reduce concentration risk at Phase 3.

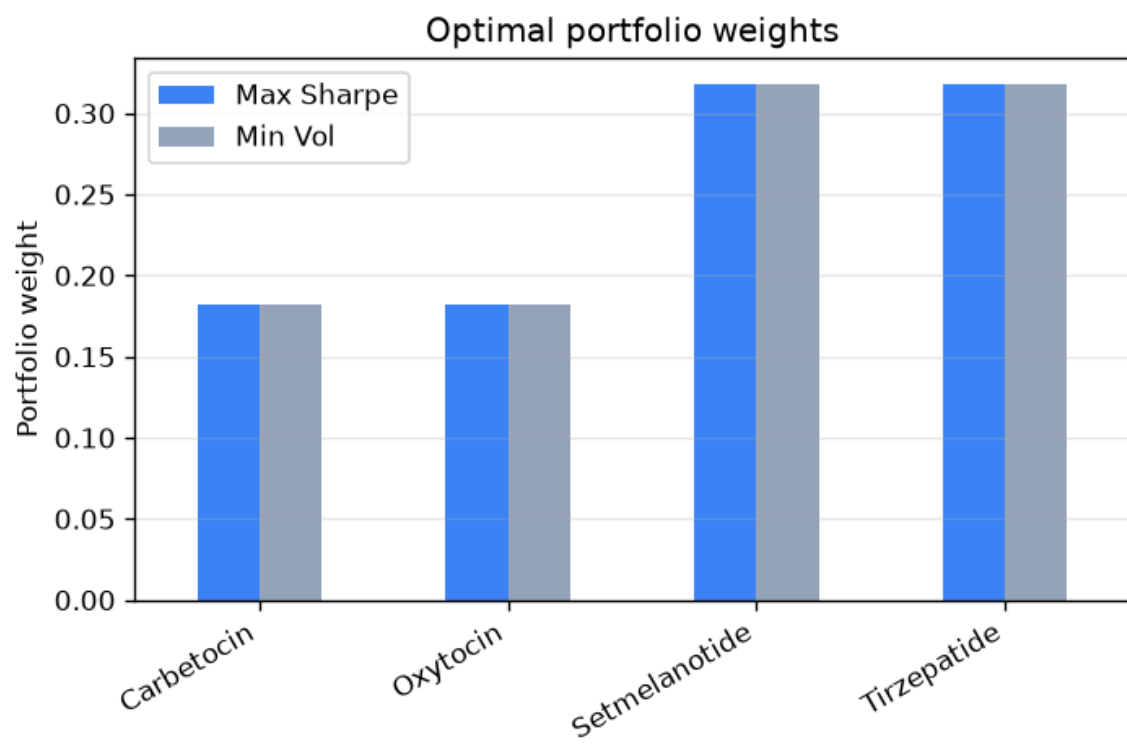
Correlation matrix

1.0 = identical bet. 0.0 = fully independent. Values above 0.5 indicate the two candidates share a biological hypothesis and should be treated as a concentrated position.



Efficient frontier & portfolio weights





Recommended allocations

Candidate	Max Sharpe	Min Volatility
Carbetocin	0.182	0.182
Oxytocin	0.182	0.182
Setmelanotide	0.318	0.318
Tirzepatide	0.318	0.318

Assumption registry

Every numeric input is listed below with its source and confidence label.

Key	Value	Conf.	Source
risk_free_rate	0.0511000000000000	high	FRED DGS10 (10-year US Treasury)
pos.phase_1	0.63	high	BIO/QLS Clinical Development Success Rates 2011-2020
pos.phase_2	0.31	high	BIO/QLS Clinical Development Success Rates 2011-2020
pos.phase_3	0.58	high	BIO/QLS Clinical Development Success Rates 2011-2020
pos.nda_bla	0.85	high	BIO/QLS Clinical Development Success Rates 2011-2020
cost.phase_0_usd	1000000	medium	DiMasi 2016 (J Health Econ), Wouters 2020 (JAMA)
cost.phase_1_usd	10000000	medium	DiMasi 2016 (J Health Econ), Wouters 2020 (JAMA)
cost.phase_1_2_usd	15000000	medium	DiMasi 2016 (J Health Econ), Wouters 2020 (JAMA)
cost.phase_2_usd	25000000	medium	DiMasi 2016 (J Health Econ), Wouters 2020 (JAMA)
cost.phase_2_3_usd	40000000	medium	DiMasi 2016 (J Health Econ), Wouters 2020 (JAMA)
cost.phase_3_usd	100000000	medium	DiMasi 2016 (J Health Econ), Wouters 2020 (JAMA)